

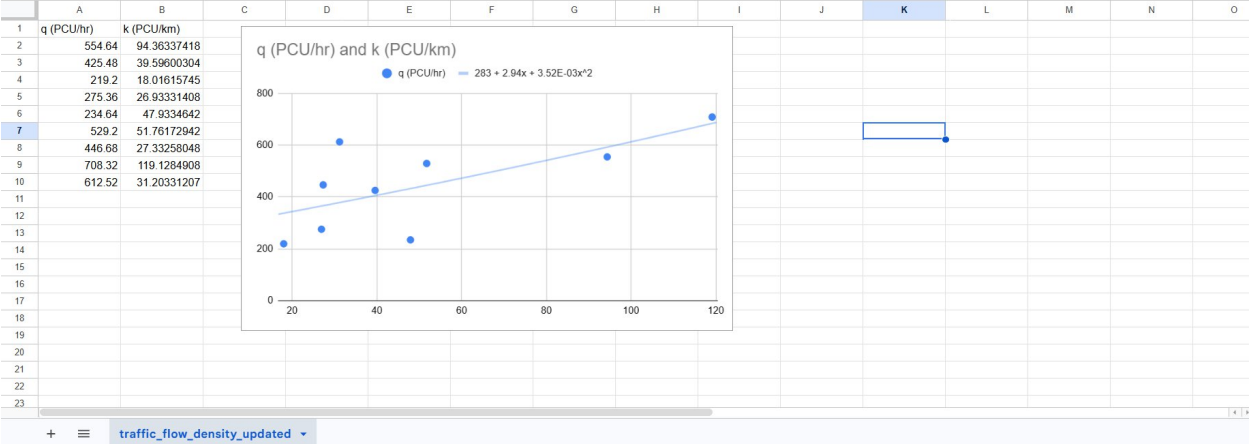




Chart editor



Setup

Customize

Chart type



Scatter chart



Data range

A1:B10



X-axis

123 k (PCU/km)



Aggregate

Series

123 q (PCU/hr)



Add Series



Switch rows / columns



Use row 1 as headers



Use column B as labels



Treat labels as text



Chart editor



Setup

Customize



Chart style



Chart & axis titles



Series



q (PCU/hr)



Format

Color



Fill color



Auto



Fill opacity

100%



Point size

7px



Point shape



Circle



Axis

Left axis



 Chart editor ×

Setup

Customize

Left axis ▼

Format data point Add

☐ Error bars

☐ Data labels

☒ Trendline

Type

Polynomial ▼

Line color

 ▼

Polynomial degree

2 ▼

Line opacity

40% ▼

Line thickness

2px ▼

Label

Use Equation ▼

☐ Show R^2

TRAFFIC MODELING CONCEPTS: SCATTER CHARTS & LOS

Part 1: How to Create the Scatter Chart in Google Sheets

- **Highlight your data:** Select the data columns for Flow (q) and Density (k), such as columns A1 to B10.
- **Insert the chart:** Go to the top menu and click "Insert" and then select "Chart".
- **Choose Scatter Chart:** In the Chart Editor panel on the right, change the "Chart type" to "Scatter chart".
- **Set the X-Axis:** Click on

the X-axis box and select Density (k) from the dropdown list. • **Set the Y-Axis (Series):** Go to the "Series" section below it, click "Add Series", and choose Flow (q). • **Check the settings boxes:** Make sure the checkboxes at the bottom are checked for "Use row 1 as headers" and "Use column G as labels". • **Add the Trendline:**

- Click on the "Customize" tab at the top right of the editor panel.
- Click to open the "Series" section.
- Scroll down and check the "Trendline" box.
- Change the trendline Type to "Polynomial" and keep the Degree at 2.
- Change the Label dropdown setting to "Use Equation" to show the math formula on your chart.

Part 2: Why We Use a Scatter Chart • Independent data points: A scatter chart treats every observation as a separate, unconnected pair of numbers rather than connecting them chronologically like a line chart. • **Find patterns:** It helps us see how data naturally groups together, spot unusual road behavior, and draw a trendline through the dots. • **X-Axis = Density (k):** This is the independent variable that represents how packed the street is with vehicles. • **Y-Axis = Flow (q):** This is the dependent variable that shows how many vehicles pass by per hour, which changes based on how packed the street is.

Part 3: Understanding the Traffic Equation • The Formula: The equation is: $q = 283 + 2.94x + 0.00352x^2$ (where x represents Density). • **An Upward Curve:** Standard highways curve downward, but our urban street data curves upward. • **Mixed Traffic Resilience:** This upward curve proves that mixed traffic (rickshaws, cars, vans) is highly resilient and keeps moving even when extremely crowded. • **The Math Floor (283 PCU/hr):** The number 283 is the baseline constant of the formula. • **Negative Limits:** If you enter a real-world flow lower than 283, the formula fails and calculates an impossible negative density. • **Why this is useful:** This shows that a single "one-size-fits-all" equation cannot model quiet, low-traffic times, meaning we need separate formulas for each street intersection.

Part 4: Real-World Data vs. Model Predictions • Real-World Density (k): The actual vehicle counts measured directly on the streets of Luxmibazar. • **Model Density (x1):** The theoretical density predicted by our Python script based only on traffic flow. • **Node A (Steady and Predictable):** The model works very well here because Node A behaves normally. For example, when real flow is 425.48, the model predicts a density of 45.94, which is very close to the real density of 39.60. • **Node C (Highly Volatile):** The model fails during quiet morning hours because the street is extremely efficient. For a morning flow of 612.52, the real density is only 31.20, but the model blindly predicts a crowded density of 100.09, resulting in a massive error of 68.89.

Part 5: Empirical Reality vs. Standard Highway Theory • Trendline Direction: Standard theory uses a downward curve, while our mixed aggregate data uses an upward curve due to combining three unique streets. • **Flow Collapse:** Standard theory says traffic collapses when density passes 60–70 PCU/km, but our streets hit their peak flow of 708.32 PCU/hr at an extreme density of 119.13 PCU/km. • **Gridlock Trigger:** Standard theory triggers "Level of

Service F" (failure) when volume collapses, but our streets trigger it when speed drops to walking pace (5.94 km/h) while volume stays very high. • **Low-Traffic Scaling:** Standard models scale down smoothly to zero, but our model hits a math floor at 283 PCU/hr and creates negative density errors.

Part 6: Level of Service (LOS) Framework Tiers • Tier 1: Uncongested Regime (LOS A to C):

- Occurs when traffic is very light, speeds are high, and roads are mostly empty.
- Flow is less than 300 PCU/hr, density is under 30 PCU/km, and speeds are 10 to 19 km/h.
- This falls below our math model's floor of 283 PCU/hr. • **Tier 2: Transition Peak Regime (LOS D to E):**
- A highly efficient state where vehicles fill the lanes but keep moving smoothly because different vehicles share space well.
- Flow is 400 to 600 PCU/hr, density is 30 to 60 PCU/km, and speeds are 10 to 16 km/h. • **Tier 3: Urban Forced Creep Regime (LOS F):**
- Extreme congestion where the street is completely packed.
- Instead of a total standstill, mixed traffic continuously squeezes and creeps forward at a walking pace.
- Flow is greater than 550 PCU/hr, density is over 90 PCU/km, and speed is less than 6 km/h.